

Grade 7 Accelerated
Unit 3
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

Rewrite each expression as an equivalent exponential expression with positive exponents.

1. $\left(\frac{3}{4}\right)^{-3} = \left(\frac{4}{3}\right)^3$

2. $\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^2$

3. $\left(\frac{-4}{5}\right)^{-4} = \left(\frac{5}{-4}\right)^4$

4. $\left(\frac{22}{7}\right)^{-7} = \left(\frac{7}{22}\right)^7$

5. Match each expression in Column A to an equivalent expression in Column B.

Column A	Column B
$\left(-\frac{11}{10}\right)^{-3}$	$\left(-\frac{11}{10}\right)^3$
$\left(\frac{11}{10}\right)^{-3}$	$\left(\frac{10}{11}\right)^3$
$\left(-\frac{10}{11}\right)^{-3}$	$\left(\frac{11}{10}\right)^3$
$\left(\frac{10}{11}\right)^{-3}$	$\left(-\frac{10}{11}\right)^3$

6. Generate an equivalent expression using the laws of exponents. Express your answer using positive exponents.

$$\left(-\frac{14}{17}\right)^{-2} \cdot \left(-\frac{14}{17}\right)^{-4}$$

$$\left(-\frac{17}{14}\right)^2 \cdot \left(-\frac{17}{14}\right)^4 \quad \left(-\frac{17}{14}\right)^{2+4} = \left(-\frac{17}{14}\right)^6$$

Consider the expression $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$. For each expression in questions 6-10, determine if it is equivalent to $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$. Then, justify your choice by explaining using laws of exponents or showing your work.

Expression	Is It Equivalent to $\frac{\left(\frac{1}{2}\right)^3}{2^{-6}} \cdot 2^6$?	Justify or Explain
7. $\frac{\left(\frac{1}{2}\right)^3}{\left(\frac{1}{2}\right)^6} \cdot 2^6$	Yes	$\left(\frac{1}{2}\right)^{-3} \cdot 2^6 = 2^3 \cdot 2^6 = 2^9$
8. 2^{15}	No	$2^{15} \neq 2^9$
9. $\frac{(2)^3}{(2)^6} \cdot 2^6$	No	$2^{-3} \cdot 2^6 = 2^3$
10. $\frac{(2)^{-3}}{(2)^{-6}} \cdot 2^6$	Yes	$2^{-3+6} \cdot 2^6 = 2^3 \cdot 2^6 = 2^9$